Judes Tumuhairwe

Staff Software Engineer | Tech Lead

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Professional Summary

■ Tech Lead with 15+ years experience cloud-native fault-tolerant, high availability, low latency distributed systems. Expert in Java-based microservices, cloud-native architecture with a proven track record of building 0-to-1 prototypes while mentoring engineers to achieve operational excellence.
■ A force multiplier with a passion for mentoring junior engineers, sets examples of good code and best practices and helps other ICs contribute to their full potential
■ Sole committer and maintainer of Experiment4J - a developer library that enables safe refactoring of critical paths by creating experiments that define exit and entry criteria as prerequisite to migration.
■ tumuhairwe.github.io

Experience

Senior Software Engineer — Aug 2024 – Present

BILL — San Jose, CA

The Invoice Financing (IF) team at BILL is responsible for short-term collateral borrowing where small businesses use outstanding customer invoices to secure immediate cash advances.
Impact: We (the IF team) bring in about $2.4 to $2.6 million every quarter in net new revenue.
My Role

- **System Ownership & Modernization**: Drive a multi-year architectural roadmap to decompose a 15-year-old, revenue-critical monolith into a high-throughput microservices architecture, supporting a working capital ecosystem generating $2.4M–$2.6M in net new quarterly revenue
- **High-Throughput Operations**: Design and deploy critical features, including the Eligibility Checker for Identity Verification and the Risk Assessment engine, serving as the core gatekeeper for the short-term collateral borrowing lifecycle.
- **Observability & Health**: Own the complete software lifecycle from initial design through production support, proactively monitoring microservice telemetry using Splunk and tracing performance anomalies via Tableau data visualization systems.
- **Technical Leadership**: Lead technical design discussion, write design docs and enforce a high bar for code quality through rigorous peer reviews
- **API & Service Design**: Implement business-critical backend functionalities utilizing Apollo GraphQL APIs routed to a performant Micronaut and Java 17 service tier

Environment

- Micronaut, Gradle, Oracle, GraphQL, SonarQube, Docker, Kafka, Redis, AWS Splunk

Tech Lead — Jul 2022 – Nov 2023

Capital One — McLean, VA

As the Tech Lead for the Internal Digital Recoveries(IDR) teams in the Card Tech org I was responsible for leading 3 different vendor teams to bring recoveries in-house (instead of being sold to debt-buyers).
Impact: The IDR project lead to $208 million in net new revenue.
My Role:

- **Distributed Systems Leadership**: Led 3 cross-functional, geographically distributed engineering teams across multiple time zones in the design, deployment, and operational maintenance of highly available, fault-tolerant microservices using Spring Boot, Kafka, and AWS (RDS, Lambda).

- **Strict Operational Ownership**: Maintained live-service operations on a rotational, on-call basis, leveraging New Relic, Datadog, CloudWatch, and Splunk to achieve rapid real-time incident resolution and maintain uptime postures for high-throughput financial transactions.
- **High-Stakes Technical Governance**: Authored and defended mission-critical technical design papers before the enterprise Design Review Board, ensuring recovery-system architectures adhered to Privacy by Design principles and strict data governance standards.
- **Organizational Leverage & Alignment**: Facilitated strategic alignment between cross-functional squads (Card Tech, Legal, and Finance) to smoothly migrate core recovery infrastructure in-house, replacing legacy external vendor systems with a centralized, scalable platform that generated $208M in net new revenue.

Environment
Spring Boot, Gradle, Postgres, Angular, SonarQube, Datadog, Docker, Kafka, MongoDB, SAFe, AWS (EC2, RDS, Lambda, CloudWatch), New Relic, Splunk, Snowflake, Java 21

Senior Software Engineer — Oct 2019 – Apr 2022

Ford Motor Company — Dearborn, MI

My team architected and operated the Connected Vehicle Data Operating System (CVDOS), an internal developer platform that enables Ford engineers to build, test and deploy vehicle commands at scale. We engineered the tool that reduced the ideation-to-prototype lifecycle by 65% for internal engineering customers.
Impact: The CVDOS project reduced time from ideation to prototype by 65% for engineers at Ford.
My Role:

- **Developer Platform Architecture:** Developed cloud-native Spring Boot microservices using Kafka to orchestrate asynchronous message dispatching and response re-aggregation across global vehicle fleets, leading to a 65% reduction in ideation time.
- **SDK Integration & Data Ingestion:** Led integration of the Microsoft SDK to consume high-throughput telemetry and command responses from vehicles transforming raw signal into actionable datasets for internal customers.
- **Engineering Velocity & Mentorship**: Created foundational platform onboarding documentation and directly mentored junior engineers on optimizing codebases for non-functional requirements, including system latency, scalability, and code maintainability.
- **Cross-Team Influence & Blameless Culture**: Fostered a rigorous culture of system reliability by spearheading cross-functional code reviews and leading blameless post-incident retrospectives to maintain strict enterprise quality standards.

Environment
Java 17, Spring Boot, SQL Server, Angular, Splunk, Kafka, SonarQube, Redis, Pivotal Cloud Foundry, Google Cloud Platform, PostMan, Git

Software Engineer — Apr 2017 – Sep 2019

Cengage Learning — Farmington Hills, MI

My team was responsible for building the subscription platform (“Cengage Unlimited”) to counter dwindling textbook sales by making thousands of books at the "Netflix of publishing” available in various formats.
Impact: Cengage Unlimited was the first in the country to introduce a subscription-based model to publishing which directly increased recurring revenue.
My Role:

- **High-Scale Platform Integration**: Developed core Identity and Access Management (IAM) and secure Single Sign-On (SSO) infrastructure for "Cengage Unlimited" (a subscription platform supporting 1.5M+ active student users a.k.a the "Netflix of Publishing"), managing member authentication while strictly prioritizing consumer privacy and data standards..
- **Fault-Tolerant Architecture**: Led technical team retrospectives to identify and resolve system bottlenecks, leveraging Hystrix and RxJava to implement resilient circuit breakers and non-blocking execution flows during high-traffic back-to-school peaks.

- **Technical & Agile Leadership**: Mentored junior software engineers in adopting production best practices, championing the integration of comprehensive unit and integration test suites, automated CI/CD workflows, and regular post-mortems.
- **Full-Stack Collaboration**: Partnered closely with UX/UI designers, product owners, and QA engineers to scale a unified student dashboard, securely caching and serving thousands of concurrent multi-format digital assets (eBooks, video, and audio streams).

Environment
Java 8, Spring Boot, RxJava, Hystrix, Maven, Bower, AngularJS, Splunk, Jenkins

Software Engineer
Feb 2016 – Apr 2017

Advaita Bioinformatics
Plymouth, MI

- My team was responsible for building the tools used scrape data from public repositories, parse it and upload it into the search index.

Outcome: We created the 1st one-stop shop for genomic data in the bioinformatics industry

■ I developed cloud native microservices, writing scripts to collect, parse and upload data from multiple public repositories and improving the user-experience.
■ I developed and integrated UI controls with multiple java microservices.
■ I wrote bash and perl scripts to download, parse and merge data from multiple public repositories and integrate it into our search engine index and other applications.

Environment
Agile, Spring Boot, Hibernate, Zuul, Eureka, S3, SNS, SQS, RDS, Maven, Grunt, Bower, AngularJS

Software Engineer
Sep 2014 – Dec 2015

Ithaka/JSTOR
Ann Arbor, MI

- My team ("Content Ingest") was responsible for receiving content (zip files) from publishers of books and journals and itemizing, characterizing, and preparing them for use by other teams (e.g. search-index team) and to build the UI publishers use to upload and track new journals/issues as they move through different states

Impact: With Content Ingest publishers can quickly upload new editions of journals and make them searchable by all universities, which increased revenue for both Ithaka and the publishers

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  ■ **High-Throughput Ingestion:** I was responsible for monitoring clusters, and running the jars on Hadoop clusters to run transforms on large datasets. I implemented microservices deployed as REST APIs to access metadata about content.
  ■ **Service Discovery and Scaling:** Proposed and implemented code and architectural changes to move away from Hadoop to REST-apis. Enabled service-location using Eureka.
  ■ **State Machine Implementation:** Implemented tracking of content as it went through different phases using finite state machine
  ■ Participated in PagerDuty rotation to help customers understand issues, triage and often resolve them.

Environment
Spring, Cassandra, Hadoop, Swagger, Amazon S3, SNS, SQS, Apache Solr

Software Engineer
Aug 2012 – Sep 2014

Domino's Pizza
Ann Arbor, MI

- My team ("Platform team") built and scaled the global ordering platform backend, ensuring 99.99% availability for a multi-client ecosystem (Web, iOS, Android) during peak global traffic events.
- **Impact**: Scaled the Platform so it could support more front-ends, leading to more orders and increased revenue e.g. serving millions of concurrent users during Superbowl

My Role

- **Extreme Scale & High Availability**: I performance-tuned the core platform to support millions of concurrent users during the Superbowl, managing massive traffic spikes through optimized caching and distributed resource management.

- **Global Architecture & Internationalization**: I spearheaded the internationalization of the backend infrastructure, re-designing data storage and user identity schemas to support rapid market expansion while adhering to diverse regional data residency and localized privacy laws.
- **Latency Optimization & Resilience**: Implemented multi-level caching strategies using EhCache and utilized Gatling/JMeter for rigorous load testing to identify and eliminate system bottlenecks, ensuring a low-latency, personalized ordering experience.
- **System Decoupling & Orchestration:** Decomposed monolithic ordering logic into resilient microservices and integrated Mule Enterprise Service Bus (ESB) for sophisticated service orchestration, message routing, and data transformation.
- **Fault-Tolerant Design:** Engineered localized menu and pricing modules to ensure system reliability across multiple locales, maintaining a consistent high-availability posture during complex global rollouts.

Environment
Spring, JMeter, Jenkins, SoapUI, Oracle, Mule ESB, Cucumber, Gatling, MuleSoft

Software Engineer
Jan 2008 – Jul 2012

Ford Motor Company
Dearborn, MI

- Project 1:
- Part Library 2 is a repository of all the parts used on all the vehicle-programs in Ford PL 2 eased recalls and has saved the company $7 million annually data integrity gave the designers and business a single view of the data

My Role

- ■ Managed the 'journey' of millions of data records through custom ETL processes, ensuring data integrity and building robust ingestion gatekeepers for high-volume upstream sources.
  ■ Designed and built batch jobs to periodically fetch data from upstream external systems
  ■ Interacted with vendor APIs to extract data, transform it based on dynamic logic, and load (or ignore) it into the new system. Also wrote the UI for users to request ad-hoc batch jobs
  ■ Designed a notification system (email, pager) to be invoked when certain data becomes available

Project 2
Automated Certification System (ACS) is a system to automate the certification of new engines by the EPA. Before, tests and certification took a year but with ACS, it now takes 3 months enabling faster turnaround, improvement in failing designs and faster time to market.

- My Role

Modernized core data infrastructure by migrating legacy systems to high-efficiency Java services designed for automated, large-scale data processing and validation.
Project 3:

- Cost Reduction Idea Database (CRID2) is a system to automate the collection of ideas from engineers, suppliers, business managers, and partners on how to reduce costs anywhere in the production chain (from conceptualization, design, production to advertising and marketing) across all vehicle programs structured way that can be quickly evaluated, rated, approved and applied.
- Project 3:
  ■ Implemented a finite state machine to process business logic as data goes from submitted to approved/integrated for various models
  ■ Designed the data model in concert with the data architect

Environment
Agile, J2EE, IBM Websphere, Oracle 10g®, MS SQL Server, IBM Message Queue®, IBM RSA®

Education

Master of Science (M.S.) in Applied Information Systems
May 2006

Towson University
Towson, MD, US

Bachelor of Science (B.S.) in Computer Information Systems
May 2004
Towson, MD, US

Towson University

Bachelor of Science (B.S.) in Business Administration
May 2004
Towson, MD, US

Towson University

Skills

- **AI & Productivity Tools:** GitHub Copilot, Gemini, n8n, GitLab Duo
- **Architectural Patterns:** API Design for State Management (GraphQL & RESTful APIs), Modular Design.
- **Modernize Frontend/Backend:** Backend (Spring Boot/Java) and Frontend (Typescript, Angular)
- **Technical**: Docker & Kubernetes, Spring, Kafka, SQL, NoSQL & Databases
- **Monitoring**: New Relic, Datadog & Splunk
- **Other**: Tech Debt Management, API Design, Mentoring and coaching

Certifications

- AWS Certified Solutions Architect
- AWS Certified Developer

Publications

- Cloud computing is not just for Fortune 500, 06/02/2023, Judes Tumuhairwe, https://minoritybusinessreview.com/cloud-computing-is-not- just-for-fortune-500

Open Source Leadership

Experiment4J (Sole Maintainer & Committer)

- **Ecosystem Stewardship**: Forked and revitalized an abandoned framework, modernizing the codebase to serve as a robust developer library for safe production refactoring and monolithic migrations.
- **Paper-Driven Feature Evolution**: Actively spearheading the project roadmap by implementing core testing and isolation features adopted from GitHub's original Ruby implementation papers for the Scientist pattern.
- **Safe Code Experimentation**: Engineered shared abstractions that allow teams to execute legacy and candidate code paths asynchronously, establishing strict entry/exit criteria to validate critical paths with zero user-facing degradation.
- **Telemetry Integration**: Built automated real-time performance tracking and telemetry reporting directly into InfluxDB to safely analyze execution behaviors during massive architectural shifts

- Active on several mailing-lists including Mule ESB, JBoss Web Services, Netbeans, Apache Axis, Open JPA, etc.

Databases

Oracle, MySQL, SQL Server, Postgres
Snowflake, Cassandra
Redshift, DynamoDB, Redis, MongoDB